Geometric Refinement and Structural Post-Processing for Scene-Level Articulated Part Understanding

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Abstract

We present a diagnosis-driven solution to the Articulate3D challenge at OpenSUN3D, ECCV 2026. Starting from released USDNet checkpoints, we improve scene-level movable-part segmentation and articulation prediction through three stages. First, we freeze the segmentation pathway, train an 88K-parameter attention-pooled origin head, and apply geometric rules for translation axes and rotation origins. Second, offline DBSCAN splitting, fragment filtering and non-maximum suppression separate merged instances. Together these stages improve the movable-task test joint AP from 0.2640 to 0.3760. Third, full-network mirror-flip finetuning improves the final movable score to **0.4056**; a three-model interaction ensemble reaches mask AP_{50} **0.3286** from a 0.2686 baseline. Our entries finished first on both dev leaderboards and second on both test leaderboards.

The report contributes geometric refinement rules, a structural post-processing recipe, and an analysis of training and evaluation pitfalls, including BatchNorm-buffer drift, restarted coarse-to-fine supervision, and floating-point failures in axis matching. We distinguish the frozen-model recipe from the later full-network finetunes and separate the intermediate two-model interaction ensemble (0.3240) from the final three-model result. These are retrospective competition experiments under a limited submission budget, rather than controlled, multi-seed ablations. Validation is in-sample for the released checkpoints and the later train-plus-validation finetunes; its gains cannot establish held-out generalization or a memorization mechanism.

1 Introduction

Articulated part understanding asks a model to find every movable part in a scene-level 3D scan and to predict how it moves: the motion type (rotation or translation), the motion axis, and — for rotations — the origin of that axis. The Articulate3D benchmark [2] poses this task on 279 released challenge scenes drawn from high-fidelity ScanNet++ [6] indoor scans, with the USDNet baseline built on Mask3D [4]/MinkowskiNet [1].

Our starting point is a faithful reproduction of the USDNet baseline (test $AP_{50}^{\text{axis+origin}} = 0.2640$, matching the officially published 0.26). We do not change the baseline architecture in the initial frozen-model stages. Later experiments finetune the full network. Throughout, we follow a *diagnosis-first* methodology: measure exactly where the metric loses points, attribute each loss to a mechanism in the pipeline, and prefer the cheapest intervention — ideally a training-free geometric one — that addresses that mechanism.

Our contributions are (1) mask-conditioned geometric refinement of axes and origins, (2) structural splitting of merged predictions across both tasks, and (3) an empirical account of freezing, curriculum and evaluation pitfalls. The following observations motivate these components.

1. **Articulation was the initial bottleneck.** Our mask AP_{50} (0.437) already led the public leaderboard. The deficit relative to the leading entry was entirely in *articulation retention* — the fraction of correctly segmented instances that also pass the axis and origin criteria (65% for us vs. 77% for the leader at the time).

2. **The two articulation heads fail differently.** Axis errors are bimodal: a large correct mode plus a 90° failure mode. Origin errors are unimodal but systematically biased. A single fix cannot address both.

3. **Scene geometry already contains most of the answer.** Doors and drawers are planar, hinges live on part boundaries, and handles sit opposite hinges. Reading these facts off the predicted masks outperforms the learned regressors that were trained to approximate them.

4. **Instance merging is a structural defect, and structural fixes transfer.** The model routinely merges the two leaves of a double door (or two adjacent handles) into one instance, corrupting every geometric quantity computed on the merged mask. Splitting masks into spatially connected components is a structural intervention; on the movable task its validation gain transferred substantially to test (§7).

5. **Validation and test gains can differ substantially.** Both the released checkpoints and the later augmented finetunes use train+validation. The interaction finetune improved validation by +0.004 and test by +0.027 (§8). This difference is descriptive evidence from this campaign, not a controlled demonstration that memorization caused the discrepancy.

2 Related Work

Scene-level articulation. Articulate3D [2] extends part-level articulation prediction from object-centric datasets to full ScanNet++ scenes. Its reference method is USDNet: a Mask3D-style transformer instance segmenter with added MLP heads that regress a 3-vector axis and origin per query.

Canonicalized articulation estimation. ANCSH [3] showed that predicting joint parameters in a normalized canonical space outperforms direct regression in camera space. RPMArt [5] votes joint parameters from point tuples for robustness. Our PCA-normal and endpoint priors follow the same philosophy — express the target in a frame the part itself defines — but require no training at all.

Test-time geometric reasoning. Our priors can be seen as a deterministic, interpretable post-process, in contrast to learned refinement stages. They leave the segmentation branch untouched by construction.

3 Preliminaries

This section fixes the task, the metric, and the data. It also states exactly which organizer-provided artifacts we consumed — a point that matters later, because two of them were trained on the validation split.

3.1 Task and metric

For each scene the submission provides N instances, each with a point mask, a confidence, a class $c \in \{\text{rotation}, \text{translation}\}$, an axis $\mathbf{a} \in \mathbb{R}^3$ and an origin $\mathbf{o} \in \mathbb{R}^3$. The leaderboard metric — we call it the *joint metric* and write it $\text{AP}_{50}^{\text{axis+origin}}$; the toolkit calls it `MAO_ST` — counts a prediction as true positive iff

$$\text{IoU}(\text{mask}) > 0.5, \qquad \arccos\big(|\cos(\mathbf{a}, \mathbf{a}^{gt})|\big) < 15^\circ, \tag{1}$$

$$\big\|P^{\perp}_{\mathbf{a}^{gt}}(\mathbf{o} - \mathbf{o}^{gt})\big\| < 0.25\,\text{m} \quad \wedge \quad \big\|P^{\perp}_{\mathbf{a}}(\mathbf{o} - \mathbf{o}^{gt})\big\| < 0.25\,\text{m}, \tag{2}$$

where $P^{\perp}$ projects out the component along an axis; translation instances skip the origin test.

Three properties matter in practice. First, the axis test is *sign-agnostic*. Second, origin correctness is a *point-to-line* distance, so sliding an origin along the axis is free. Third, the evaluator's projection divides by $\|\mathbf{a}\|$ rather than $\|\mathbf{a}\|^2$, so predicted axes must be normalized — an implementation detail worth $+0.018$ $\text{AP}_{50}^{\text{axis+origin}}$ on validation by itself.

3.2 Data: what the organizers provide, and what we consumed

Everything in this paper runs on organizer-provided artifacts; we used no external data. The Articulate3D annotations cover 279 ScanNet++ v1 indoor scenes with 6349 articulations (4687 rotation, 1662 translation). They comprise: movable-part instance masks with per-part articulation parameters (type, axis, origin) for the main task; interaction-element instance masks (handles, knobs, buttons) for the companion task; and a USD scene-hierarchy annotation, which we did not use. The official split is 195 train / 42 validation / 42 test scenes. Test labels are withheld and scored only by the challenge server. The organizers additionally release preprocessed point clouds (the mesh downsampled at a 1 cm voxel size; evaluation runs on 2 cm-voxelized clouds carrying the ground truth) and two USDNet checkpoints, `mov_trainval.ckpt` and `inter_trainval.ckpt`. Both checkpoints were trained on train+validation — a fact that matters twice in what follows.

How each piece was used, concretely:

- **train (195 scenes):** the only data our early head finetune (the 88K origin attention head, §5.2) trained on.
- **train+validation (237 scenes):** what the released checkpoints saw — the root of the in-sample validation caveat discussed (§8) — and what our augmented finetunes `v5`, `v6`, `v5b` and `mov_v11` retrain on. Their validation results are consequently also in-sample.
- **validation (42 scenes):** our offline arbiter. We rebuilt the official evaluation to bit-exactness against the challenge server (verified on two submissions), so all iteration ran offline without spending submission quota. Every diagnosis statistic in this paper — error forensics, oracle ceilings, cue inventories — is measured here.
- **test (42 scenes):** points only, no labels. Every test number in this paper is a server-returned score of a numbered submission.
- **Cross-task coupling:** predicted interaction masks are an *input* to the movable task via the hinge-opposite-handle rule (§5.4). Companion-task quality therefore propagates into main-task origins, which §7 exploits deliberately.
- **Per-scene arrays:** all offline post-processing, including the DBSCAN splitting of §7, consumes only the preprocessed per-scene `.npy` arrays (xyz, RGB, normals, semantic and instance ids) — no meshes, no images, no re-inference.

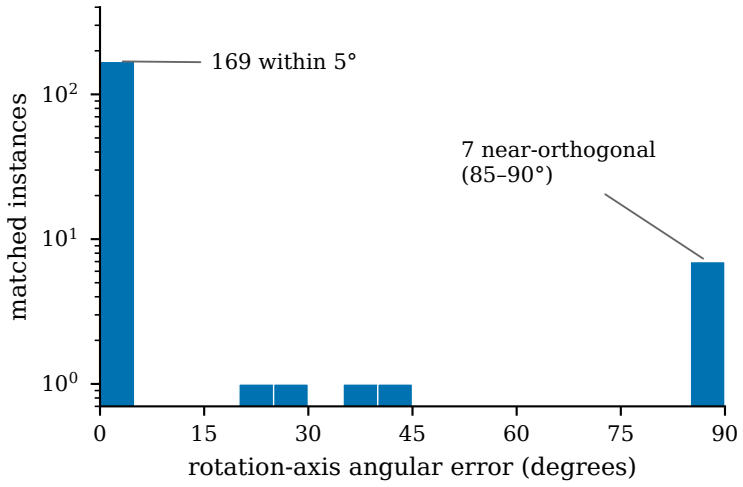


Figure 1: **Baseline rotation-axis errors** (log-y; $n = 180$ matched validation rotations): 169 instances fall within 5° (median 2.7°), while the 7 lateral hinges all fail at 85–90° — the head outputs $\approx (0, 0, 1)$ regardless of the part.

Not used: the ScanNet++ RGB streams (DSLR or iPhone), meshes beyond the preprocessed clouds, the USD hierarchy/connectivity annotations, and any external data. Our hand-crafted RGB and normal cues did not show useful hinge-side discrimination in the measured validation probes (§6.5); this does not establish that these modalities lack information for other models or feature constructions.

3.3 Baseline

USDNet predicts articulation from the transformer queries: per-point features inside each predicted mask are *mean-pooled* and decoded by MLPs into axis and origin (its "mode 2"). We reproduce the released checkpoint at test $\mathrm{AP}_{50}^{\text{axis+origin}} = 0.2640$.

4 Diagnosis

Before changing anything, we measure where the baseline loses points. The two heads fail through different, identifiable mechanisms, which licenses fixing them separately.

We match predictions to ground truth on the validation split (greedy, IoU > 0.5; 275 matched instances) and analyze the two heads separately. Our harness reproduces the official validation numbers bit-for-bit (and matched the challenge server on a probe submission), so all analysis is against the exact deployed metric.

Axis: mode collapse onto gravity. 93.3% of matched rotation axes in GT are vertical within 15°, and the regression head, trained with a $1 - \cos$ loss on this distribution, collapses onto that mode: every lateral hinge (oven/dishwasher-style) fails at 85–90° (Figure 1). Strikingly, the trivial rule "always predict gravity" passes 93.3% of rotation axes vs. the model's 88.3% (0 wins / 9 losses for the model). The learned head's entire value lies in translation azimuths, where gravity is always wrong.

Origin: centroid shrinkage from mean aggregation. Rotation-origin error is unimodal (median 0.127 m against a 0.25 m threshold) with no global offset: an oracle global translation

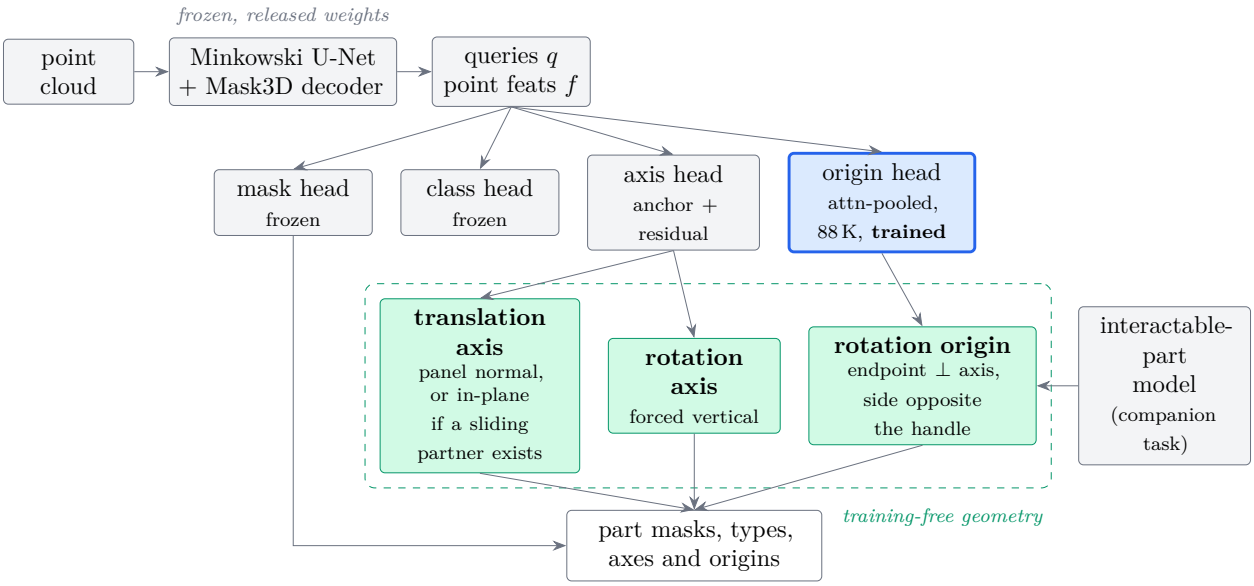


Figure 2: **The recipe.** Everything on the mask pathway is frozen at the released weights, so AP_{50} is bit-identical to the baseline by construction. Exactly one component is trained — an 88 K-parameter attention-pooled origin head, with BatchNorm statistics pinned (§5.1). The articulation parameters that the metric actually scores are then *overwritten* by training-free geometry read off the predicted mask: the axis head's own output survives only as the anchor prior inside the dashed box. Handle masks for the hinge-side rule come from the challenge's companion interactable-part task.

improves the pass rate by only +0.006. The real bias is *shrinkage*. 77% of predicted origins are displaced from the GT hinge toward the part centroid, recovering only 29% of the centroid-to-hinge distance. The mechanism is architectural: mode-2 aggregation averages (offset + coordinate) over *all* mask points, and a mean over a door pulls any estimate toward the middle of the door, while hinges live on its edge. Consistently, an oracle that snaps origins to mask endpoints given GT axes recovers the hinge for 100% of instances (median residual 8 mm), while the centroid passes only 28%.

Independence. Axis and origin failures are nearly independent ($\phi = -0.11$) in the matched validation sample, motivating separate interventions without implying that their AP gains must add.

5 Method

Figure 2 depicts the initial frozen-model recipe, not the later full-network finetuning stage. The evaluator recomputes joint AP using the conjunction of mask, axis and origin matching criteria; products of retention factors elsewhere in this report are interpretive heuristics, not a definition of the official joint AP.

This section describes that recipe. The design follows directly from the diagnosis: keep the segmentation frozen, train one small origin head, and overwrite the articulation parameters with geometry read off the predicted masks.

Checkpoint (same weights)	AP_{50}	AP_{50}^{axis}	AP_{50}^{origin}	$AP_{50}^{axis+origin}$
baseline	0.5711	0.4419	0.4495	0.3215
+ head, BN *drifted*	0.5720	0.4558	0.4513	0.3369
+ head, BN *restored*	0.5711	0.4419	0.4486	0.3215

Table 1: **BatchNorm-drift confound (validation).** A 4-epoch trained origin head appears to help (+0.015) — until the drifted BN buffers are restored to the source checkpoint, revealing the head itself contributed nothing. All 124 BN buffers had moved; all frozen parameters were bit-identical.

5.1 Freezing discipline: keep what already works

Because segmentation led the leaderboard, our first design rule is that the mask pathway must remain *bit-identical*. Freezing parameters (`requires_grad=False`) is **not sufficient**: BatchNorm running statistics are buffers, refreshed by any forward pass in training mode. In an early experiment all 124 BN buffers drifted while every non-articulation parameter stayed frozen, which changed the masks and manufactured a +0.015 validation "gain" that vanished when the statistics were restored (Table 1). We therefore pin every `_BatchNorm` module to `eval()` for the whole run, re-applied each epoch, since the trainer resets module modes.

5.2 The one trained component: attention-pooled origins

We replace the mean over mask points with a learned attention pooling: 88K parameters, a scoring MLP plus the original origin MLP. It trains with everything else frozen as in §5.1, so any change in $AP_{50}^{axis+origin}$ is attributable to the aggregation alone. Training uses the metric-aligned point-to-line loss and a sign-agnostic $1 - |\cos|$ axis loss — the metric takes $|\cos|$, and penalizing a 180° flip with maximum loss fights the evaluator. The head trains on the 195 train scenes only. The released baseline was trained on train+val. An observed validation difference of ∼0.01 between checkpoint variants is not a clean estimate of the effect of training on validation, because the training histories also differ.

5.3 Axis anchors instead of free regression

Following the bimodality in §4, the axis is expressed as a choice between two *anchors*: gravity $\mathbf{z}$, and the free regression axis flattened into the horizontal plane. We keep the regressed azimuth rather than quantizing it; a fixed azimuth grid measurably loses (0.449/0.466/0.468 AP_{50}^{axis} for $\{z{+}4\}/\{z{+}8\}$ azimuths/ours). With zero-initialized selection and residual layers the module *exactly* reproduces "snap to the nearest anchor" at load time. That alone is worth +0.024 AP_{50}^{axis} on validation with no training.

5.4 Training-free geometric priors

This subsection gives the three rules that overwrite the learned articulation outputs — translation axis, rotation origin, hinge side — each targeting one mechanism from §4.

(i) Translation axis from panel geometry. Sliding parts are planar, and the panel's own geometry localises the slide direction far better than the network does. The smallest-PCA direction of the predicted mask's points, flattened to the horizontal plane, has median azimuth error 0.70° on matched validation translations; the axis head's is 5.54°. We adopt the PCA direction whenever it

agrees with the model's axis within 25°; the agreement gate protects non-planar parts. Because the metric skips the origin test for translations, this directly lifts the translation half of $\mathrm{AP}_{50}^{\text{axis+origin}}$.

A planar panel admits *two* horizontal slide directions, however, and assuming the normal is a statement about drawers only. A sliding door or window leaf travels *along* its own face, so its axis is the in-plane horizontal — exactly 90° from the normal. Nine matched validation translations are of this kind, and the rule above gets every one of them maximally wrong. The model cannot arbitrate: having learnt a drawers-slide-outward prior, its axis lies closer to the normal on all of them.

We therefore decide the question geometrically, from the partner a sliding leaf must have. Two leaves sit in *different* planes about one leaf thickness apart, so that one can pass behind the other, and they stand beside each other at matching height. A row or stack of drawers shares one plane exactly, and coplanar fronts cannot pass each other. Concretely, a translation part is flipped to its in-plane direction when another predicted translation panel is parallel within 20°, offset through the plane by 0.004–0.06 m, displaced laterally, overlapping in height, and itself planar and of matching height. On validation the plane offset separates the two classes without overlap: the in-plane cases sit at 0.008–0.035 m, the drawer-like ones at 0.000, 0.089, 0.245, 1.30 and 1.84 m.

One further condition is invisible on validation: a sliding leaf must be door-sized. A row of drawer fronts is also same-height and near-coplanar, so the pair test alone accepts it, and on the test split the rule was rotating 0.23 m and 0.33 m panels — one of them the highest-scoring prediction in its scene — until we added a 0.45 m floor on leaf height. Every true in-plane case on validation is at least 0.50 m tall, so the floor leaves the validation score bit-identical. A validation sweep cannot discover this kind of condition: the split simply contains no counterexample.

The rule's validation gain does not transfer in proportion, and the reason is visible before submitting: it fires 91 times on the 41 validation scenes but only 35 times on the 42 test scenes, because eight of its nine validation positives come from one wardrobe. The +0.021 it earns there measures how much sliding furniture that split contains. Plateau-checking guards against a threshold being fitted to the data; nothing guards against the *phenomenon* being concentrated in it.

Requiring the partner to be planar and height-matched — rather than testing only whether the part itself is flat — is what makes this safe. The one-sided version peaks higher at a planarity threshold of 0.05 but falls *below* the no-flip baseline at 0.03 and 0.08; eight of its nine positives come from a single validation scene, so it is fitted to one piece of furniture. Stating the pair symmetrically flattens the surface: with a height tolerance of 0.15, every planarity threshold from 0.03 to 0.35 improves $\mathrm{AP}_{50}^{\text{axis+origin}}$, by between +0.004 and +0.021.

(ii) Rotation origin = perpendicular endpoint. We snap each rotation origin to one of the two robust (5/95% quantile) endpoints of the mask along its widest direction *perpendicular to the predicted axis*. The perpendicularity is essential. A door's longest direction is vertical, and snapping along the first PCA direction moves origins to the top/bottom edge, costing −0.19 $\mathrm{AP}_{50}^{\text{origin}}$; the correct construction recovers +0.014.

(iii) Hinge side = opposite the handle. Which of the two endpoints carries the hinge? Doors open away from their handles. We reuse the challenge's companion task: the interactable-part model predicts handle masks, and if a predicted handle lies near the part, we choose the endpoint *farther* from it, falling back to the endpoint nearest the predicted origin otherwise. On matched validation rotations this raises the origin pass rate from 78.3% to 82.8%. Notably it picks the "correct side" slightly *less* often (86.1% vs. 88.9%): its mistakes concentrate on parts where both endpoints fail anyway, while its wins occur where the choice matters.

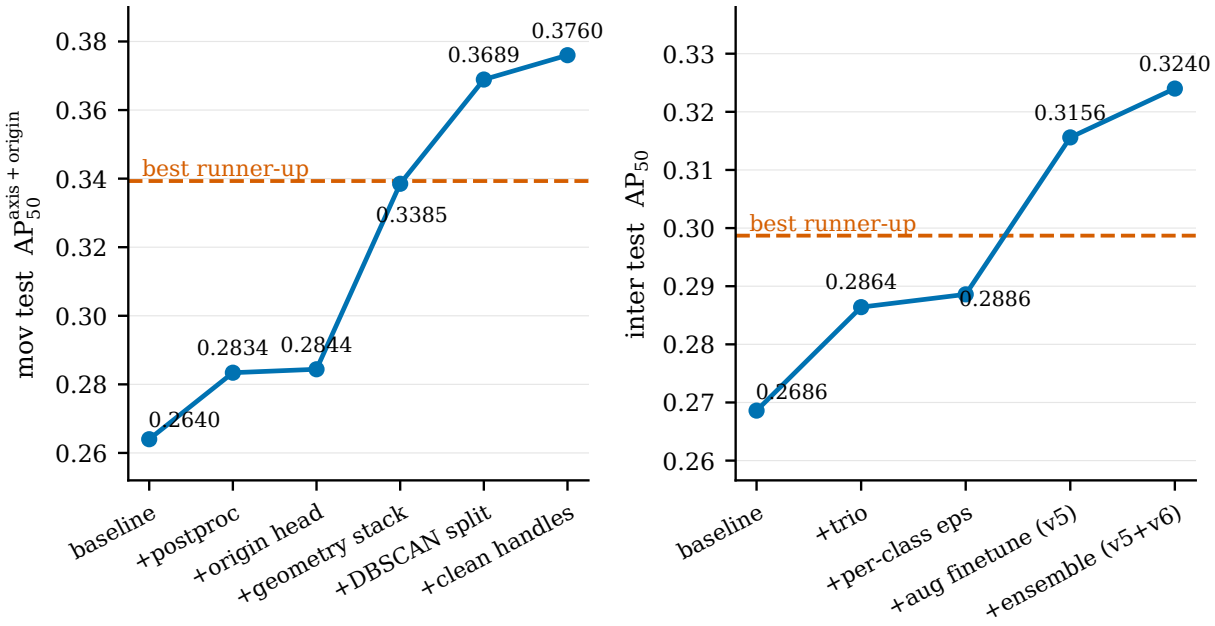


Figure 3: **Official test-score trajectories**; every point is a server-returned score. Left, movable task: 0.2640 (baseline) → 0.3385 (geometric recipe) → 0.3689 (DBSCAN split) → 0.3760 (cleaned handle source). Right, interaction task: 0.2686 → 0.2886 (per-class trio) → 0.3156 (`v5` finetune) → 0.3240 (ensemble). Dashed lines mark the best visible competitor at the original deadline (0.3393 and 0.2904). A deadline-day retrain later raised our final scores to 0.4056/0.3286 (§8).

6 Experiments

This section reports the main results, then four studies of the deployed pipeline: a re-diagnosis, a confounded submission and its controlled replacement, the remaining bottleneck, and the negative results.

6.1 Setup

Articulate3D challenge splits: 195 train / 42 val / 42 test scenes. All validation numbers use the official evaluator; our local harness matches the challenge server bit-for-bit (verified on two submissions). Inference is nondeterministic (unseeded); repeated runs vary by ±0.002, which we treat as the noise floor.

In the initial recipe and its post-processing studies, the backbone, decoder and mask head retain the released USDNet weights; the full-network finetunes of §8 are an explicit exception. Thus, so AP_{50} is constant across post-processing changes *within* a checkpoint. It is not constant across checkpoints. Our train+val origin retrain leaves the predicted masks byte-identical but alters `pred_classes` and `pred_scores`; because average precision depends on the confidence ranking and the per-class grouping, measured AP_{50} moves from 0.4370 to 0.4366. We flag this because we initially assumed otherwise and reported a carried-over AP_{50} for the retrained rows.

Configuration (cumulative)	AP_{50}	AP_{50}^{axis}	AP_{50}^{origin}	$AP_{50}^{axis+origin}$
Validation				
USDNet baseline (reproduced)	0.5711	0.4419	0.4495	0.3215
+ trained origin attention	0.5711	0.4419	0.4717	0.3477
+ axis anchors (warm start)	0.5716	0.4677	0.4717	0.3743
+ PCA panel normal (§5.4.i)	0.5716	0.4836	0.4717	0.4049
+ endpoint + handle side (§5.4.ii–iii)	0.5716	0.4836	0.4926	0.4306
+ swept endpoint quantile & handle margin (§6.3)	0.5716	0.4836	0.5018	0.4395
+ sliding-leaf axis flip (§5.4.i)	0.5716	0.5046	0.5018	0.4605
+ rotation axes forced vertical	0.5716	0.5063	0.5018	0.4621
+ DBSCAN structural split (§7)	0.5801	—	0.5393	0.5021
+ cleaned handle source (§7)	0.5801	—	0.5411	**0.5038**
Test (challenge server)				
USDNet baseline (reproduced)	0.4370	0.3801	0.3160	0.2640
val-fitted post-processing only	0.4370	0.3887	0.3254	0.2834
full recipe	0.4370	0.4060	0.3601	0.3345
+ train+val origin retrain (2 runs)	—	—	0.357	0.331–0.332
+ sliding-leaf flip (confounded, §6.4)	0.4366	0.4051	0.3578	0.3322
+ leaf flip + vertical rotation axes	0.4370	0.4101	0.3601	0.3385
+ DBSCAN structural split (§7)	0.4444	0.4105	0.3955	0.3689
+ **cleaned handle source (best pre-extension)**	**0.4444**	0.4105	**0.4026**	**0.3760**
Leaderboard leader (concurrent, pre-extension)	0.4352	**0.4135**	0.3547	0.3360
Runner-up (pre-extension board)	—	—	—	0.3393

Table 2: **Main results.** All rows above the DBSCAN split keep the segmentation branch frozen; AP_{50} is constant *within* a checkpoint, but not across them (the train+val retrain leaves masks byte-identical while altering `pred_classes` and `pred_scores`, which moves AP_{50} to 0.4366). The leaf-flip row stood first on the ranking metric at the time, ahead of the concurrent leader by 0.0024; it differs from the *full recipe* row only in translation and rotation axes — AP_{50} and AP_{50}^{origin} return bit-identical to seven decimals, so its +0.0041 axis gain is attributable to those two rules alone (§6.4). The DBSCAN structural split is the single deliberate mask-touching operation before full-network finetuning (§7); the cleaned-handle row is our best pre-extension score, 0.0367 ahead of the pre-extension runner-up. The deadline-day retrain that supersedes it appears in §8 and Table 3. Validation axis sub-metrics for the split rows were not re-measured in isolation (—).

6.2 Main results

Table 2 summarizes the progression; Figure 3 traces the server-verified trajectories on both tasks. Figure 4 shows the pattern that organised the campaign: how much of a validation gain reaches test depends on the *kind* of change. Validation-tuned thresholds transferred at roughly 1:3 — double optimism from fitted hyperparameters plus a validation set used to train the released backbone. These effects are not isolated by the comparisons. The geometric recipe transferred at ∼1.2:1 (val +0.042 → test +0.049): panel normals and hinge-opposite-handle are properties of the world, not of the split.

A third regime is neither of these. The sliding-leaf rule is parameter-robust and physically correct, yet transferred at only 1:5.5 (val +0.0227 → test +0.0041), because the phenomenon it fires on is concentrated in the validation split (§5.4). A rule's *firing count on the target split* is observable before submitting and predicts the realised gain far better than its validation gain does; we recommend reporting it.

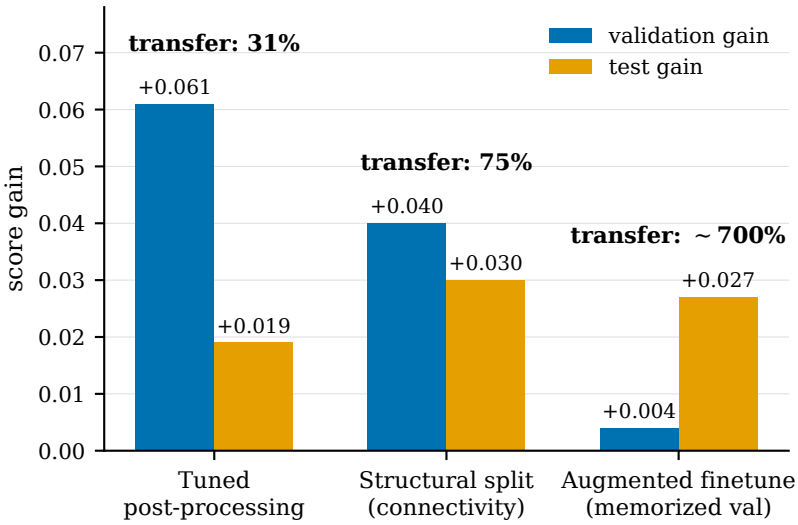


Figure 4: **Validation gain vs. realised test gain, by kind of change.** Tuned thresholds keep 31% of their validation gain on test; the structural DBSCAN split keeps 75%; the mirror-flip finetune realises ∼700%. These ratios describe observed changes; both the baseline and finetune validation scores are in-sample, and the ratios do not identify a causal mechanism.

6.3 Re-diagnosing the deployed pipeline

Every diagnosis in §4 was run on *baseline* predictions. Once the geometric priors rewrite every axis and every rotation origin, the error profile is a different distribution. We initially failed to re-measure it, and spent considerable effort on lateral hinges (7 instances not recovered by the methods tested) while the actual residual sat elsewhere. Re-running the per-instance breakdown on the deployed stack:

	axis pass	origin pass	both
rotation (n=180)	93.3%	83.3%	80.6%
translation (n=95)	91.6%	—	91.6%

Rotation failures decompose as 5 axis-only, **23 origin-only**, 7 both. After the geometric axis work, origin dominates the residual by 4.6×. Six instances sit within 1.5× of a threshold: two translation axes at 15.2°/16.8°, and four origins at 0.306–0.340 m.

That points at two hand-picked constants never swept. The endpoint quantile is strongly unimodal with a broad plateau, and the handle-ownership margin is monotone:

quantile	2/98	5/95	**10/90**	12/88	15/85	20/80	25/75	30/70
$AP_{50}^{axis+origin}$	.4187	.4306	**.4359**	.4354	.4345	.4292	.3810	.3027
handle margin (m)	0.10	0.15	0.25	0.35	0.50			
$AP_{50}^{axis+origin}$	**.4395**	**.4395**	.4359	.4359	.4330			

Both curves are plateaus rather than spikes, which is the criterion we used to accept them. The mechanisms are interpretable: past the ∼20th percentile the "endpoint" is no longer an endpoint (hence the collapse), and a generous ownership margin lets a *neighbouring* part's handle vote and

flip the hinge side. Two further constants swept in the same run — the canonical-snap cone (15°–45°) and the handle score cut (0.3–0.7) — are completely flat, i.e. saturated. Together these lift validation 0.4306 → 0.4395.

6.4 A confounded submission, and the controlled one that replaced it

We first packaged the sliding-leaf rule on top of a different checkpoint than our best submission used, changing the base and the post-processing at once. The server returned 0.3322 against a standing best of 0.3345, which reads as the rule having hurt. It had not: that checkpoint's own baseline scored 0.331–0.332, so the rule was flat-to-positive and the loss came from the base. The two changes were unseparable after the fact, and the submission slot was wasted.

We then re-packaged the rule as a *minimal edit*: the identical prediction file with only translation axes rewritten. This made the next result readable, and let us state a falsifiable prediction beforehand — AP_{50} and $\mathrm{AP}_{50}^{\mathrm{origin}}$ must come back bit-identical. They did, to seven decimal places (0.4369668 and 0.3601476), while $\mathrm{AP}_{50}^{\mathrm{axis}}$ rose by 0.0041007 and $\mathrm{AP}_{50}^{\mathrm{axis+origin}}$ by 0.0039516. That number is the two axis rules' contribution with nothing else mixed in. The resulting submission stood first on the ranking metric at the time: 0.33845 against the previous leader's 0.33601.

The discipline generalises: when submissions are scarce, change one thing and predict an invariant that must survive it — the invariant tells you the measurement is clean before you interpret it.

6.5 The remaining bottleneck is selection, not representation

This subsection examines the measured residual: origins remain a weak factor on the matched validation subset. Its predicted masks contain near-hinge points, but the tested selectors do not outperform the handle rule reliably. These findings are conditional on this subset and model.

Pushing past our submitted score means improving one of the three factors in $\mathrm{AP}_{50}^{\mathrm{axis+origin}} \approx \mathrm{AP}_{50} \times r_{\mathrm{axis}} \times r_{\mathrm{origin}}$, which on test read $0.4370 \times 93.85\% \times 82.42\%$. The arithmetic points somewhere counter-intuitive. Making the axis *perfect* yields only 0.360; making the origin perfect yields 0.410. Origin, not axis, is the weak factor — and we had spent our effort on the axis.

Origin turns out not to be limited by segmentation at all. For every matched rotation the predicted mask *contains* a point within 0.25 m of the ground-truth axis line, median 6 mm: the mask ceiling is 100%. Against that, the deployed rule scores 85.0%, the best of its two endpoint candidates 88.3%, and a widened family sweeping quantile and depth 95.6%. The answer is present and we cannot select it.

We then asked which signal could. We treat the hinge *edge* as the single latent that determines both quantities: the axis runs along it, and the origin lies on it. For each available cue we measured its own AUC for "is this the hinge edge", sampling along the whole edge rather than in a ball around one endpoint:

signal	AUC	individual accuracy
handle distance	**0.820** (12.2σ)	**90.4%**
emptiness beyond the edge	0.567 (2.0σ)	57.1%
normal crease at the edge	0.519	53.8%
colour difference across the edge	0.493	48.7%
contact with static geometry	0.449	37.8%
colour variance along the edge	0.435	35.9%

The tested RGB and surface-normal features did not provide evidence of useful discrimination. Handle distance was the strongest measured cue and was already deployed. These negative probes do not rule out other features or nonlinear models using the same inputs. This also explains a failure we had found puzzling: a logistic selector over these features, under leave-one-scene-out CV, scores 78.9%, *below* the hand-written rule, because it dilutes the one real signal with five noisy ones.

Finally we asked whether the network's own representation knows better than our hand-crafted cues — the question behind "train a new head rather than retrain the backbone". We probe the 256-d input to the anchor module (query embedding concatenated with mask-pooled point features) over 137 instances, leave-one-scene-out, against a permutation null. Predicting where along its width the hinge sits gives $R^2 = -0.301$ against a null of -0.213 ± 0.115, i.e. 0.8σ — no useful signal detected by this probe. With a frozen backbone a new head must exploit the frozen features. This probe did not recover hinge location reliably; it does not establish information-theoretic absence.

This corrects our own earlier reasoning: we had leaned for days on a probe with only *seven* positives — far too few for an AUC to establish absence. The larger probe also failed to find useful predictive performance, but its size, choice of readout and uncertainty still limit the conclusion.

6.6 Summary of negative results

We tested alternative axis heads, origin selectors, mask snapping and augmentation strategies. Detailed outcomes and failure analyses are retained in Appendix A. They are configuration-specific observations, not proofs that the broader research directions cannot work.

6.7 An evaluator artifact: bit-exact axes are scored as misses

This subsection documents a floating-point artifact in the official evaluator that makes *perfect* predictions fail. It silently caps every oracle, and it inverts the usual assumption that an oracle measures the headroom of a component.

The axis criterion is implemented as

```
deriavation = np.dot(axis_ref, axis_checked) /
              (np.linalg.norm(axis_ref) * np.linalg.norm(axis_checked))
is_axis = np.arccos(np.abs(deriavation)) < np.deg2rad(15)
```

For two *bit-identical* unit vectors the quotient does not evaluate to 1.0: the roundings of the numerator and of the product of norms disagree, and the result is 1+1 ULP = 1.0000000000000002. Then arccos returns `NaN`, and `NaN < th` is `False`. An exactly correct axis is recorded as a miss. Sampling generic unit vectors, 36.4% of self-comparisons overflow this way. Axes with exactly representable components, such as the canonical $(0, 0, \pm 1)$, always give exactly 1.0 and are safe.

That distinction is what makes the artifact so asymmetric here. Of the 236 validation rotation axes, 206 are stored as canonical $(0, 0, \pm 1)$ (the benchmark is gravity-dominated, cf. §4), so their NaN rate is only 4.2%. None of the 154 translation axes is canonical, and their rate is 50.6%. Consequently the ideal oracle — GT masks, GT classes, GT axes and GT origins, all confidences 1.0 — scores $\mathrm{AP}_{50}^{\text{axis+origin}} = 0.6516$ rather than 1.0, with mask $\mathrm{AP}_{50} = 1.0000$ and $\mathrm{AP}_{50}^{\text{origin}} = 1.0000$. The entire deficit is in the axis term, and 90.7% of it falls on translations. The evaluator applies the criterion in both matching directions, so each overflow costs a hard false negative *and* a false positive (88/390 instances fail in the GT loop, 90/390 in the prediction loop; the asymmetry is the rounding of $\mathbf{a}\cdot\mathbf{b}$ versus $\mathbf{b}\cdot\mathbf{a}$). The resulting per-class arithmetic closes exactly, e.g. for rotations $\frac{1}{2} \cdot \frac{226}{236} \cdot (1 + \frac{226}{237}) = 0.9354$, matching the evaluator's reported 0.9354.

Two consequences are worth stating plainly. First, **being exactly right is punished**: pasting the true GT axes onto our model's own predictions *lowers* AP_{50}^{axis} from 0.4432 to 0.3860. Second, **the score depends on serialization precision**. Rounding the same GT axes before submission changes nothing geometrically but moves the oracle to 0.9951 at `%.6f` (the format our own submission writer uses), 0.9743 at `%.7f`, 1.0000 at five decimals or fewer, and 0.7368 when cast to `float32` — because rounding perturbs the vector just enough that the quotient lands strictly below 1. A one-line `np.clip(deriavation, -1.0, 1.0)` in a copy of the evaluator restores the ideal oracle to exactly 1.0000 on all four sub-metrics, confirming that this single line accounts for 100% of the gap.

We verified that nothing else contributes. No GT instance is dropped by `min_region_sizes` (smallest is 16 points) or by the valid-class filter. GT masks are disjoint by construction (one integer per point, and mask AP_{50} is exactly 1.0). No GT axis is degenerate or flagged `needsReview`. Point counts agree across the annotation, the processed tensors and the prediction pickle for all 42 scenes, and the AP integration itself returns exactly 1.0 for an all-true-positive curve at any confidence assignment. All measurements and the intervention are reproducible via `analysis/oracle_gap.py`; the original evaluator was left untouched.

6.8 The axis head has collapsed completely

Re-diagnosing the deployed pipeline shows the gravity collapse is not partial but total. 95.6% of the 2854 predicted rotation axes are *exactly* $[0, 0, 1]$, and printing the remaining 4.4% against ground truth shows they are simply wrong. Forcing them all vertical is therefore free, and worth +0.0017 AP_{50}^{axis} at zero parameters.

The consequence is that rotation-axis accuracy is now a property of the data rather than of the model. It equals the fraction of ground-truth rotation axes lying within 15° of vertical: 169/180 = 93.9% on validation. Of the twelve failures before the snap, seven are lateral hinges (ground truth horizontal); four have genuinely tilted ground truth (23.4°, 29.7°, 35.0°, 44.7° from vertical — so no snapping rule can reach them either); one was the single non-vertical prediction, which the snap fixes. With translation at 95.8%, articulation post-processing has little measured headroom under the tested rules; this is not an upper bound over all possible post-processors. The geometric detector, the learned selector, and rebalanced training all failed in instructive ways (§A). Mask recall (70.5% of GT instances matched at IoU 0.5) caps $AP_{50} = 0.571$ on validation and is the largest untouched quantity. Our TTA experiments show it cannot be recovered at inference time from a backbone trained without the corresponding augmentation; unlocking it requires retraining the segmentation branch with the (now fixed) geometric augmentations, which we deliberately kept out of scope to preserve the leading mask quality. Events later tested both halves of that judgment: an offline structural split found mask gains that post-processing could still reach (§7), and a deadline extension funded precisely the augmentation retrain deferred here — first on the interaction task, then on the movable model itself, and it paid both times (§8).

One untested direction addresses the collapse at its documented cause. An axis is a point on $\mathbb{RP}^2$, not a free vector in $\mathbb{R}^3$, and our treatment of it is Euclidean throughout: per-point votes are averaged linearly (so opposite signs cancel, one plausible route into a single dominant mode), anchor residuals are added and renormalised rather than pushed through an exponential map, and anchor selection is a discrete softmax — the same softmax whose confidence wall we showed makes the lateral-hinge classifier immovable. Replacing the discrete selector with continuous regression in the tangent space at an anchor removes that wall by construction, and aggregating with the leading eigenvector of $\sum_i a_i a_i^\top$ is the sign-correct mean on $\mathbb{RP}^2$. Taken further, a rotation axis and its origin jointly specify a *line* in $\mathbb{R}^3$, whose natural home is the affine Grassmannian under Plücker

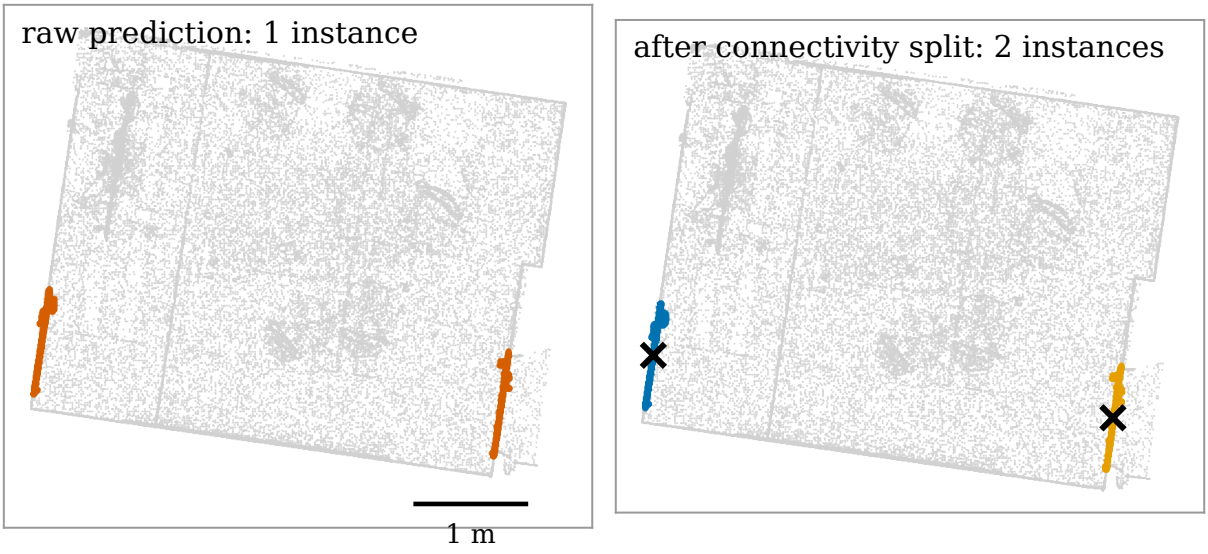


Figure 5: **Structural splitting, validation scene `5748ce6f01`.** Left: the model merges two distant, visually similar door leaves — on opposite walls, 4.2 m apart — into one predicted instance, so merging is a structural defect of the model, not a proximity artifact. Right: DBSCAN ε-connectivity separates the mask into its two spatial components (5267 and 5295 points; centroids marked), and every downstream geometric quantity is then computed per leaf.

coordinates — which is also how the evaluator scores them, as a point-to-line distance. We flag this as the one remaining idea aimed at a known failure mechanism rather than at a symptom.

7 Structural Post-Processing: Two Doors Are Two Objects

This section covers the campaign's second stage: an offline splitting of merged instances that lifted both tasks, and the one deliberate exception to the frozen-mask rule.

§6.8 concluded that articulation post-processing was exhausted, and the frozen-mask rule of §5.1 was our guarantee of clean attribution. Forty-eight hours before the original deadline, looking at per-scene predictions rather than per-instance statistics surfaced a failure class no threshold sweep would have found. A disclaimer up front, applying to this section and the next: with one submission per day on each test leaderboard, these results are iterative competition engineering — one variable at a time wherever quota allowed, server-verified wherever a slot existed — not controlled ablations, and we flag every comparison that is confounded.

The companion interaction task now enters the story as more than a handle source, so we restate it (cf. §3.2): it segments interactable elements (handles, knobs, buttons), scored by plain mask AP_{50} with no articulation term. Until this point our entry there was the released interaction checkpoint, unmodified: validation 0.5363, test 0.2686.

7.1 The failure: a merged mask corrupts everything downstream

The model routinely merges the two leaves of a double door — or two adjacent handles — into a *single* predicted instance (Figure 5). The direct cost is familiar: against either ground-truth instance the merged mask has IoU ≈ 0.5, so both matches sit at the metric's threshold and usually

fall below it. The indirect cost is worse and specific to our pipeline: every geometric prior of §5.4 is computed *on the mask*, so a merged mask's PCA axis runs diagonally across the pair and its "perpendicular endpoint" lands in the gap between the doors. A merged instance does not merely lose its own point; it poisons the axis and origin that the whole method reads off the geometry.

7.2 The DBSCAN trio: split, filter, suppress

The fix is offline connected-component splitting. Run DBSCAN (`min_samples`= 1, so it is pure ε-connectivity) on the points of each predicted mask, emit each spatial component as its own instance, drop components below a minimum size, and finish with IoU-based NMS to remove duplicates. We call this three-step post-process the *trio*. It runs on the dumped predictions plus the preprocessed point coordinates — no GPU, no re-inference.

For the movable task ($\varepsilon = 0.2\,\text{m}$, minimum 50 points) the trio raises validation mask AP_{50} $0.5716 \rightarrow 0.5801$ and, rerunning the geometric stack on the split masks, the joint metric $0.4621 \rightarrow 0.5021$; on the test server, $0.3385 \rightarrow 0.3689$ (Figure 3). As the mechanism predicts, the gain concentrates in the origin term ($0.3601 \rightarrow 0.3955$): each leaf's PCA axis and endpoint origin are finally computed on the right geometry. For the interaction task ($\varepsilon = 0.075$, minimum 5 points, NMS at IoU 0.75), validation goes $0.5363 \rightarrow 0.6054$ and test $0.2686 \rightarrow 0.2864$.

Splitting addresses a structural defect, but its DBSCAN radius, minimum component size and NMS threshold are still selected on validation. On the movable task the measured transfer was substantial (Figure 4); this observation does not guarantee transfer to other datasets or splits. On the interaction task the same trio realised $+0.069 \rightarrow +0.018$; the absolute levels there differ by a factor of two (0.54 validation vs. 0.27 test), and both validation scores are in-sample (§8). We report the measured difference without attributing it uniquely to memorization.

7.3 Per-class ε: cluster radius is element scale

A single ε for the interaction task is a compromise across element sizes: handles are elongated, knobs and buttons are tiny. Splitting the radius by class — $\varepsilon = 0.1$ for handles (class 1), $\varepsilon = 0.065$ for knobs/buttons (class 2) — is one number per physical scale. It lifts validation $0.6054 \rightarrow 0.6253$ and test $0.2864 \rightarrow 0.2886$.

7.4 Cleaning the handle source

The hinge-side rule of §5.4(iii) consumes *predicted* handle masks, and those predictions carry the same merged-instance disease, so we feed the movable pipeline the trio-processed interaction predictions instead. Validation joint metric moves $0.5021 \rightarrow 0.5038$ (origin $0.5393 \rightarrow 0.5411$); on test, $0.3689 \rightarrow \mathbf{0.3760}$ (origin $0.3955 \rightarrow 0.4026$) — an over-delivery relative to the validation estimate, and our best pre-extension movable-test score. The two tasks are coupled: interaction post-processing quality is an input to movable origin accuracy.

7.5 A quota-timing mistake

One episode from the original deadline day is worth recording as a negative result about *process*. Each test leaderboard allows one submission per day. We spent the interaction-test slot at dawn on the per-class variant (0.2886). By dusk we had built a better candidate and no quota left to submit it, and a competitor's buzzer submission took first place at 0.2904. The 0.0018 deficit was not a modelling gap but a scheduling one. On a deadline day the daily submission is the day's scarcest asset, and it should be spent at the *end* of the day, on the best artifact the day produced.

8 Full-Network Finetuning and Interaction Ensembling

This section covers the final stage: full-network augmentation finetuning and interaction-model proposal ensembling. Unlike the frozen recipe, these runs may change the backbone, decoder, masks and BatchNorm buffers. An extension of the competition deadline provided additional submission slots for evaluating these candidates.

8.1 Validation protocol and its limits

The released interaction checkpoint was trained on train+validation, and so were `v5`, `v6`, `v5b` and `mov_v11`. Validation is therefore in-sample for both the baseline and these finetunes; it is incorrect to regard the finetunes as out-of-sample on that split. We nevertheless used validation for configuration selection, so it does not provide an unbiased estimate of generalization.

Augmentation was motivated by the released configuration's lack of geometric augmentation, a flip-time axis-label transform bug that we corrected, and poor recall on mirrored inputs (55.9% versus 70.5% on original inputs, §A). Test-server scores provide held-out-label measurements, but the limited, repeated submissions also informed method selection. The experiments do not isolate the contribution of memorization, augmentation, checkpoint selection or other training changes.

8.2 `v5`: the flips-only finetune

`v5` finetunes the released interaction checkpoint on train+validation with mirror-flip augmentation, coarse-to-fine curriculum *off* (§8.4), learning rate 10^{-5}, 60 epochs, batch size 1, cropping on. In-sample validation moved by +0.004 only (0.536 → 0.540). On test it moved by +0.027: 0.2886 → **0.3156**. The larger observed test gain is useful retrospective evidence for this configuration, but does not by itself identify why the validation and test changes differ. Both validation measurements are in-sample, and this comparison is not a controlled test of a memorization hypothesis.

8.3 `v6` and the equal-strength ensemble

`v6` repeats the recipe with random yaw rotation added. Solo it is weaker (0.529 validation) — but its augmentation spectrum differs from `v5`'s, which is the property an ensemble needs. We take the union of `v5` and `v6` proposals, then run the per-class DBSCAN trio over the union with NMS at 0.55: validation 0.6446, test **0.3240** — first place at the time, 0.034 clear of the then-visible runner-up's 0.2904.

We had tried proposal-union ensembling before and it failed: a union of the baseline with a weaker finetune scored *below* the baseline's own post-processed validation score (0.587 vs. 0.605) — the weak member's proposals dilute the strong member's ranking. These two experiments suggest that member strength and augmentation diversity matter for proposal unions; they do not establish a universal necessary or sufficient condition for successful ensembling.

8.4 The coarse-to-fine trap: curriculum state is part of the checkpoint

The most destructive pitfall of the campaign hides in the training schedule, not the weights. The repository's coarse-to-fine curriculum dilates the target masks by a radius $r \cdot \gamma^{\lfloor e/\alpha \rfloor}$ with $r = 0.1\,\mathrm{m}$ and $\alpha = 100$: for the first 100 epochs of *any* run, supervision is masks inflated by 10 cm. The released checkpoints were trained through the whole curriculum (dilation decayed to zero). A finetune launched from them with the curriculum enabled therefore *restarts* it, and spends its entire budget re-teaching the model to predict dilated masks.

Leaderboard (ranking metric)	ours	TnG	teamzb	margin
movable dev ($AP_{50}^{axis+origin}$)	**0.5038**	—	0.4434	+0.0604
movable test ($AP_{50}^{axis+origin}$)	0.4056	**0.4828**	0.3711	−0.0772
interaction dev (AP_{50})	**0.6446**	—	0.6394	+0.0052
interaction test (AP_{50})	0.3286	**0.3446**	0.3110	−0.0160

Table 3: **Final leaderboards** (2026-08-28): first on both dev boards, second on both test boards. Margin is ours minus the best other entry. Our final movable test sub-metrics: AP_{50} 0.4668, AP_{50}^{axis} 0.4315, AP_{50}^{origin} 0.4328.

This destroyed two runs before we found it. An interaction finetune collapsed 0.536 → 0.008 in AP_{50} while AP_{25} survived at 0.20 — the fingerprint of masks that are right but fat. A movable full-network retrain jumped in score exactly when the first decay triggered at epoch 100. A hybrid experiment (finetuned weights, baseline BatchNorm buffers) ruled out BN drift as the cause. Until the mechanism was found, these collapses had us concluding that "retraining hurts" — a wrong lesson drawn from a confounded experiment, and the reason the bet of this section was almost never placed. The general statement: *curriculum state is part of the checkpoint.* Finetuning must disable, or fast-forward, every schedule-dependent transform of the training targets.

8.5 Negative results of the extension window

The window also funded a burst of cheap experiments, most of which failed informatively; the bottom block of Table 4 lists them. One deserves prose. `v10`, a movable full-network retrain with the curriculum correctly disabled, reached mask AP_{50} 0.404 at epoch 184 against the released 0.572: fixing the trap of §8.4 was necessary but not sufficient on the movable side. The released weights carry an official 1160+680-epoch schedule, and 195 scenes at batch size 1 on a single 24 GB GPU do not close that gap in a day. The curriculum explains the interaction collapse; compute explains the movable one.

8.6 Final standings

Table 3 gives the final standings. The run in flight delivered: `mov_v11` — the `v5` recipe transplanted to the movable model (train+validation, flips only, curriculum off, learning rate 10^{-5}) — raised in-sample validation mask AP_{50} from 0.5716 to 0.5911 raw (0.6259 after splitting) and full-pipeline validation joint to 0.5319; on test, joint **0.4056** with AP_{50} 0.4668 (from 0.4444; axis 0.4315, origin 0.4328), against 0.3760 before it. The augmentation dividend transplanted to the movable model and over-delivered again. On the interaction side, a three-model ensemble (`v5`+`v6`+`v5b`; validation 0.6513) reached test **0.3286**, our final score there. The movable result uses the single `mov_v11` checkpoint plus the geometric and structural pipeline, not a two-model movable ensemble. The exact selected `mov_v11` checkpoint epoch cannot be recovered consistently from the public artifacts: the working notes record approximately 45 epochs while an earlier report stated 60. We therefore do not present an exact epoch as verified.

The final hours also redrew the boards: competitors who had kept their submissions private all month went public at the buzzer, and `TnG` took both test boards (0.4828 movable, 0.3446 interaction). The scores alone do not establish the competitors' training procedures or architectures. Their methods were not evaluated in our experiments. We finished first on both dev leaderboards and second on both test leaderboards.

9 Limitations and Reproducibility

Experimental scope. This is a retrospective competition report on one benchmark. Repeated, unseeded inference varied by approximately ±0.002; we did not run a controlled multi-seed factorial study. Cumulative results can mix checkpoint, training and post-processing changes. We identify specific confounds, but do not assign all final gains to isolated components.

Data reuse. Released checkpoints and the later full-network finetunes use train+validation. That split is in-sample, and the same labels also guide configuration selection. Test labels were withheld, but server feedback from multiple submissions informed development. Neither split supports unrestricted claims about generalization to other scenes, datasets or motion distributions.

Geometric assumptions. The recipe favors planar panels, vertical rotation axes and hinges opposite visible predicted handles. Lateral hinges, non-planar parts and incorrect interaction masks remain failure cases. Negative feature probes characterize only the measured features, samples and readouts.

Artifacts. Source, scripts and the short competition report are available at https://github.com/junyuan-fang/articulate3d. Data and checkpoints are obtained separately. Several scripts retain machine-specific paths, and the migrated Conda/Docker channel specifications have not been rebuilt on Linux/CUDA. Exact reproduction of the final movable checkpoint is limited by the unresolved epoch record described in §8. The original evaluation implementation is retained for challenge comparability; the clipped-axis diagnostic is explicitly a separate intervention.

Detailed failed configurations and additional experimental lessons are in Appendices A and B.

10 Conclusion

A reproduced USDNet baseline (0.264 test $AP_{50}^{axis+origin}$) reaches 0.4056 — +54% relative, first on both dev leaderboards and second on both test leaderboards — in three stages (Figure 3): one tiny trained head plus training-free geometric priors (0.264 → 0.3385, +28% relative); an offline structural split of merged instances (§7); and an extension-window bet — augmented finetunes on both tasks and a three-model interaction ensemble — that delivered the final scores (→ 0.4056; interaction 0.2686 → 0.3286, +22%, §8). The segmentation branch stayed frozen through the first two stages, with one deliberate, offline exception: splitting what the model had merged.

The final finetunes show that small changes on an in-sample validation split can coexist with larger test gains (+0.004 versus +0.027 for the interaction finetune). Both models trained on validation, so these numbers should not be interpreted as a controlled explanation of memorization. The axis head's total collapse onto gravity remains unsolved: it emits $[0, 0, 1]$ for 95.6% of rotations and is wrong on the rest, ten interventions failed to recover the lateral hinges, and the $\mathbb{RP}^2$ treatment sketched in §6.8 is an untested direction, not a validated remedy. We hope the more than two dozen documented negative results, the freezing, curriculum and evaluation pitfalls, and the confounded-submission postmortems of §6.4 and §8.4 are as useful as the recipe itself.

Acknowledgments. The solution builds on the USDNet/Articulate3D toolkit and the ScanNet++ dataset; we thank the challenge organizers.

A Detailed Negative Results

Table 4 lists every failed attempt of the campaign, one row each, including those of the extension window (§8). Each consumed real experiment time and shaped the final recipe. Four failures carry a mechanism instructive beyond their row; they get short postmortems below.

Attempt	Outcome	Root cause
Main campaign		
joint training, 3× oversampling + 5× loss weighting	$AP_{50}^{axis+origin}$ 0.374 → 0.089 in 4 epochs	rebalanced 13:1 →≈ 1.4:1; argmax selector mass-flips (postmortem)
bounded residual on anchored axes ($\leq$ 5.7°, residual-only trainable)	+0.004 raw; stacked 0.4219 vs 0.4306	gain absorbed by the PCA prior
wide-flat ⇒ lateral-hinge classifier	precision 8%	lateral parts include small box-like lids
unconditional panel normal (agreement gate dropped)	retention 91.6% → 96.8%, yet AP_{50}^{axis} −0.010	retention ignores the low-confidence tail (postmortem)
model arbitrates the two slide directions	wrong on 5 of 5 in-plane cases	the drawer prior is the bias being corrected
stacking detector for lateral hinges	holds for 0 of the 7 lateral parts	appliance doors do not stack as assumed
obstruction detector for lateral hinges	occupancy 0.213 vs 0.157; 3 of 7 have nothing above	premise false in this data
superpoint mask snapping (3 variants)	AP_{50} 0.5716 → 0.5699/0.5697/0.5665 (repair); 0.5122 fine / 0.2254 coarse rebuild	semantic vs. geometric boundaries (postmortem)
learned origin-side selector (§6.5)	78.9% vs the rule's 85.0%; 1.7σ from null	dilutes the one real signal
widened origin candidate family	10/90 stays optimal; face offset +0.6 pts (noise); all three side selectors exactly 85.0%	face-side sign is a cross product, not physics
seed-ensemble TTA	runs differ ±0.002; 97% shared instances, median best-IoU 1.0	no usable diversity
mirror-flip TTA (3 fusion variants)	best variant 0.4292 vs the 0.4306 single-pass stack	mirrored scenes are OOD for the backbone (postmortem)
anchor classifier for lateral hinges	+0.0002 $AP_{50}^{axis+origin}$ in 12 epochs	temperature-10 warm start: ∼10-logit wall, gradients $\mathcal{O}(10^{-5})$
"flat-lying panels hinge laterally" rule	fires on 3 parts, breaks 2; lone true lateral 90° → 58°	still a miss
"hinge edge abuts dense frame" rule	side accuracy 50.6% — a coin flip	closed doors sit flush on both sides
handle offset as translation azimuth	median angle to slide axis 76.8°	handle offsets lie in the front-face plane
geometric-confidence reranking	lowers every metric, AP_{50} included	origin displacement confounds with part size
origin-head retrain on train+val	+0.001/−0.002/−0.002 (train-only, 20-epoch, 150-epoch-budget heads)	endpoint snap overwrites the regressed origin
naive freezing (BN buffers drift, §5.1)	phantom +0.015 gain (Table 1)	buffers update despite `requires_grad=False`
Extension window (§8)		
`v4`: 300-epoch long training	peaks at 0.484, then declines	overtraining, not undertraining, binds
`v5b`: second schedule on `v5`	0.532, below `v5`	augmentation gain already banked
rescore by movable-part overlap	monotonically worse at every blend weight	handles also live on static surfaces

Attempt	Outcome	Root cause
class-confusion repair	0 of 282 well-localized predictions misclassed	nothing to fix; measured before building
interaction mask dilation	0.5659, below the 0.6054 stack	masks are not systematically small
flip-TTA on flip-trained models	val 0.6335 vs the 0.6513 ensemble	in-distribution flips add no diversity
consensus-vote mask merging	AP_{50} collapses to 0.536	majority vote biases toward intersections
`v10`: movable retrain, curriculum off	AP_{50} 0.404 at epoch 184 vs 0.572	compute gap, not curriculum (prose, §8)

Table 4: **Every failed attempt of the campaign**, one row each; the bottom block covers the extension window. Entries marked (post-mortem) are expanded below, and `v10` is discussed in §8.

Joint training with doubled rebalancing. Oversampling lateral-hinge scenes 3× *and* weighting their loss 5× shifted the effective vertical:lateral ratio from 13:1 to ≈ 1.4:1, and a hard argmax anchor selector trained on a near-balanced world mass-flips vertical hinges at deployment: $AP_{50}^{\text{axis+origin}}$ fell from 0.374 to 0.089 in four epochs. Rebalance with *one* mechanism at a time.

Dropping the agreement gate. The panel normal is eight times more accurate than the axis head, so taking it unconditionally on any flat part looks strictly better — and on matched instances it is: retention rises from 91.6% to 96.8%. End to end it *loses* 0.010 AP_{50}^{axis}, because retention is computed over high-confidence predictions greedily matched at IoU > 0.5, while AP integrates the whole precision–recall curve, including the low-confidence tail where masks are poor and panel geometry is unreliable. Separating this change from the sliding-leaf flip it had been bundled with turned a net loss into the +0.021 of §5.4. Retention is a diagnostic, not an objective.

Superpoint mask snapping. Constraining masks to unions of a geometric over-segmentation costs AP_{50} monotonically, from 0.5716 untouched down to 0.2254 for a coarse rebuild (Table 4): the limit of the intervention is doing nothing. Learned masks track *semantic* part boundaries; superpoints encode geometric coherence, and the two disagree — a cabinet door and its frame are coplanar with continuous normals, so no geometric cue separates them, while a panel's own surface may carry creases that split it. We also measured the representation ceiling only *after* the first attempt, which is backwards: at the coarse granularity the oracle over the best possible superpoint set reaches IoU ≥ 0.5 on 50.7% of instances against the 63.8% our masks already achieve, and 0.000 median on parts under 500 points. Five minutes of measurement would have skipped a collapsed experiment.

Mirror-flip TTA. The frozen backbone was trained *without* flip augmentation, so mirrored scenes are out-of-distribution: the mirrored pass recalls only 55.9% of GT instances vs. 70.5% for the original (73.8% union, i.e. only 13 genuinely new finds). Naive concatenation collapses AP_{50} (0.572 → 0.476; twins become false positives); IoU-NMS recovers most but not all (0.549); even a surgical variant that appends only non-overlapping mirrored proposals at halved score stays below the single-pass stack (0.4292 vs. 0.4306 $AP_{50}^{\text{axis+origin}}$). In this experiment, flip TTA did not recover the loss associated with the transformed input.

B Additional Experimental Lessons

We close with the campaign's transferable lessons, each tied to the experiment that taught it.

1. **Diagnose before training — and re-diagnose after every change.** Every effective component traces to a specific, quantified failure mechanism; both expensive failures (v4 collapse, BN drift) came from acting before measuring. Worse, we kept optimising against a *stale* diagnosis — days of lateral-hinge work read a map of the baseline's errors, not the deployed pipeline's; ten minutes of re-measurement (§6.3) found more.

2. **Freezing parameters does not freeze behavior.** Buffers (BatchNorm statistics) update in training mode regardless of `requires_grad`; verify claims of "frozen" by diffing checkpoints end-to-end.

3. **Match the loss to the metric.** Sign-agnostic axis loss ($1 - |\cos|$), point-to-line origin loss, and axis normalization at submission time each exist because the evaluator says so.

4. **Compare transfer empirically.** The val→test transfer ratio was $\sim 3\times$ higher for the geometric rules studied here than for validation-fitted post-processing. The practical test is not whether a threshold has an optimum but whether it has a *plateau*, and stating a rule as a relation between two parts instead of a property of one made the whole swept range safe (§5.4) — an observed plateau in this campaign, not a guarantee against dataset shift.

5. **An oracle proves existence, not reachability.** Every mask in the matched validation analysis contains a point within 0.25 m of the hinge line (median 6 mm), yielding a 100% conditional selection ceiling — and we briefly concluded that post-processing would suffice. Four subsequent methods failed to select that point. An oracle measures the representation, not whether any available signal can pick the right element out of it; report the two separately.

6. **A negative result needs power to mean anything.** We cited a seven-positive probe as proof that the backbone lacks lateral-hinge information, and let it close a research direction for days. The larger probe also found no useful predictive performance: with seven positives an AUC cannot distinguish absence from noise. Before retiring a direction on an against-the-null result, check that the null had something to reject.

7. **Measure a representation's ceiling before building on it.** This is the same discipline as the previous point, applied forwards. We ran the superpoint intervention first and the ceiling analysis after, and the ceiling turned out to be *below* what we already had.

8. **Optimise the metric you are scored on, not a proxy for it.** Per-instance retention and AP disagree in sign: taking the panel normal unconditionally raises matched-instance retention from 91.6% to 96.8% while *lowering* $\mathrm{AP}_{50}^{\mathrm{axis}}$ by 0.010, because retention sees only confident, well-matched instances and AP integrates the low-confidence tail where the same geometry is unreliable. Diagnostics that restrict to the easy subset will recommend changes that lose.

9. **Validate the oracle, not just the model.** An upper bound is a measurement of the *evaluator* as much as of the method. On this benchmark the ideal oracle scores 0.65, not 1.0, because bit-exact axes trigger an arccos domain overflow (§6.7); any headroom estimate that assumes an oracle reaches 1.0 is wrong by a third, and the error is concentrated on exactly the sub-task the method targets.

10. **Leaderboard metrics can guide hypotheses.** Retention-rate analysis of competitors' four public numbers suggested possible differences in articulation retention, and pointed to the exact sub-tasks (translation azimuth, hinge side) where the gap lay.

11. **Structural splitting transferred substantially here.** The campaign suggests three descriptive patterns of val→test changes: validation-tuned thresholds realise ∼1/3 of their gain; physically-correct rules realise it in proportion to how often they fire on the target split; and structural fixes — repairs of defects like a merged double door — transferred substantially in the movable experiments (+0.040 → +0.030 for the DBSCAN split, §7). Budget submissions accordingly.

12. **Separate observed transfer from its explanation.** The interaction finetune changed in-sample validation by +0.004 and held-out test by +0.027 (§8). Both baseline and finetune trained on validation, so this difference cannot establish a memorization effect.

13. **Ensemble composition mattered in these comparisons.** It worked with two near-equal, differently-augmented finetunes (0.3240 on test) and failed with a strong–weak pair (0.587 vs. 0.605 on validation): the weaker member degraded the union in that comparison. Check member parity before spending an ensemble submission.

14. **Curriculum state is part of the checkpoint.** Schedule-dependent target transforms (here, coarse-to-fine mask dilation restarting at $r = 0.1\,\text{m}$, $\alpha = 100$) silently reset when finetuning resumes at epoch zero, and destroyed two runs (0.536 → 0.008) before being identified (§8.4). Worse than the lost runs was the lost conclusion: the collapses read as "retraining hurts" and nearly buried the campaign's most profitable bet.

15. **On deadline days, spend the daily quota at the end of the day.** We submitted at dawn, improved by dusk, and were overtaken by a buzzer submission (0.2904 vs. our 0.2886); the 0.0018 was a scheduling gap, not a modelling one (§7).

References

[1] Christopher Choy, JunYoung Gwak, and Silvio Savarese. 4d spatio-temporal convnets: Minkowski convolutional neural networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

[2] Anna-Maria Halacheva, Yang Miao, Jan-Nico Zaech, Xi Wang, Luc Van Gool, and Danda Pani Paudel. Articulate3d: Holistic understanding of 3d scenes as universal scene description. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 5633–5644, 2025.

[3] Xiaolong Li, He Wang, Li Yi, Leonidas Guibas, A. Lynn Abbott, and Shuran Song. Category-level articulated object pose estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.

[4] Jonas Schult, Francis Engelmann, Alexander Hermans, Or Litany, Siyu Tang, and Bastian Leibe. Mask3d: Mask transformer for 3d semantic instance segmentation. In *International Conference on Robotics and Automation (ICRA)*, 2023.

[5] Junbo Wang, Wenhai Liu, Qiaojun Yu, Yang You, Liu Liu, Weiming Wang, and Cewu Lu. Rpmart: Towards robust perception and manipulation for articulated objects. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 7270–7277, 2024.

[6] Chandan Yeshwanth, Yueh-Cheng Liu, Matthias Nießner, and Angela Dai. Scannet++: A high-fidelity dataset of 3d indoor scenes. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.